Experiment 8

Configuring User Management and Role-Based Access Control (RBAC) in a Zoho Creator Application

Application: Patient Appointment Scheduling System | Zoho Creator

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Abstract

This experiment demonstrates user management and role-based access control (RBAC) for the Patient Appointment Scheduling System in Zoho Creator. The application Settings page provides permission sets and role/data-sharing controls. Existing permission sets shown in the supplied screenshot include Billing Specialist, Clinic Administrator, Doctor and Medical Receptionist, alongside the default Read and Write sets. A module-level permission set can define which forms, reports and pages a user can access and whether the user can view, edit or delete records.

1. Aim

To configure user permissions and role-based access control in the Patient Appointment Scheduling System so that each user can access only the forms, reports and records required for their responsibilities.

2. Introduction

RBAC assigns permissions according to a user's role. In a patient appointment application, the clinic administrator manages the overall system, the medical receptionist handles appointment bookings, doctors review relevant patient and appointment details, and billing specialists handle fee-related information. Permissions should follow the principle of least privilege: grant only the access required for each role.

3. Concepts

Permission sets

The Permissions tab lists named permission sets. In the supplied screenshot, the existing sets are Billing Specialist, Clinic Administrator, Doctor, Medical Receptionist, Read and Write.

Module-level permissions

The Add Permission Set dialog shows modules such as Appointment Booking Form, Doctor Master, Family Member and Dashboard, with controls for Access, View, Edit, Delete and More.

Roles and data sharing

The Roles tab is used to define role hierarchy where supported. Data Sharing controls record visibility. These settings complement the module-level permission set.

User sharing

The Share option can be used to grant application access to intended users and assign an appropriate permission set, depending on the application's available options and plan.

4. Procedure

1. Open the Patient Appointment Scheduling System in Zoho Creator and select Settings.

2. Under Permissions, open User Permissions. Review the Permissions, Roles and Data Sharing tabs and the existing permission sets.

3. Click Add Permission to open the Add Permission Set dialog.

4. Create or edit a permission set for the intended role. Use a clear name such as Medical Receptionist Access or Doctor Access.

5. In the Set Permissions table, configure access for Appointment Booking Form, Doctor Master, Family Member and Dashboard according to the user's duties. Select only the required Access, View, Edit, Delete and More rights.

6. Save the permission set. If creating separate roles, review the Roles and Data Sharing tabs and configure only the controls available in the current Creator account.

7. Use Share to grant access to an authorized test user and assign the appropriate permission set, if this option is available to your account.

8. Open the live application using the test user's access and verify which menu items and record actions are available. Confirm that restricted modules or actions are not available.

5. Suggested Permission Matrix

Role / permission set	Appointment Booking Form	Doctor Master	Family Member	Dashboard
Clinic Administrator	Manage as required	Manage as required	Manage as required	Access
Medical Receptionist	Create/view; edit only if required	View as needed	Access only if required	View
Doctor	View assigned appointments/patient details	View	View only if clinically required	View
Billing Specialist	View fee-related fields/records as required	No access unless needed	No access unless needed	View fee summaries if available
Read (default)	Read-only across components per set description	Read-only	Read-only	Read-only

Role / permission set	Appointment Booking Form	Doctor Master	Family Member	Dashboard
Write (default)	Write permission across components per set description	Write permission	Write permission	Write permission where applicable

The matrix is a recommended configuration plan, not a claim that every permission has already been applied. The exact available controls and record-level filtering depend on the Zoho Creator plan and app configuration. For real patient information, do not grant broad Read or Write access unless it is appropriate; verify record visibility and avoid exposing sensitive data to unrelated users.

6. Implementation Screenshots

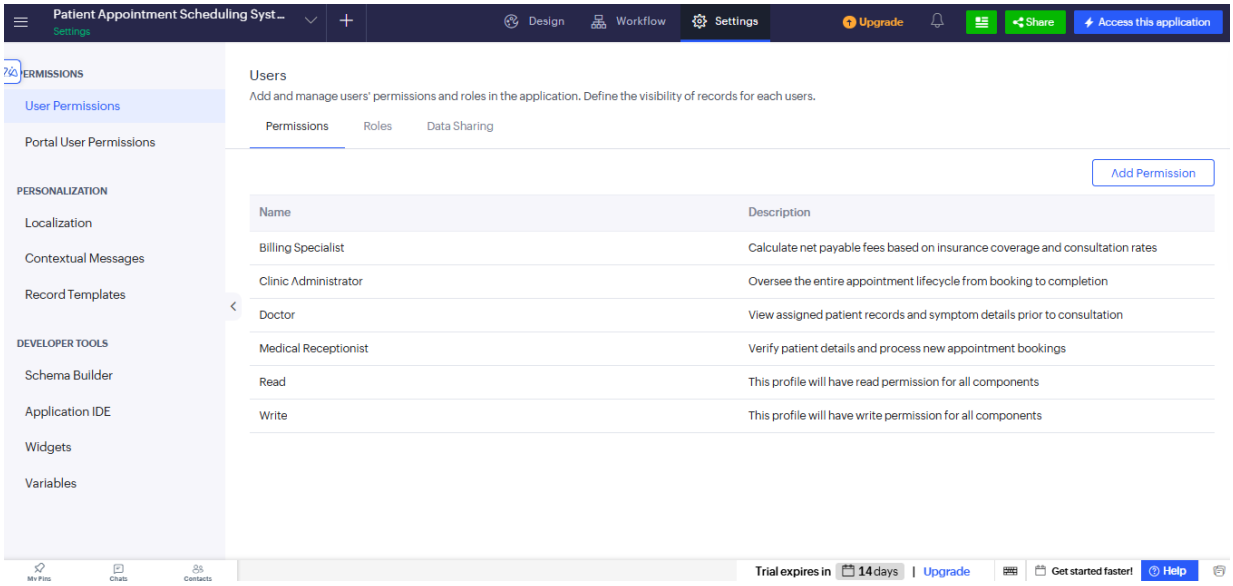


Figure 1: User Permissions page showing the available permission sets.

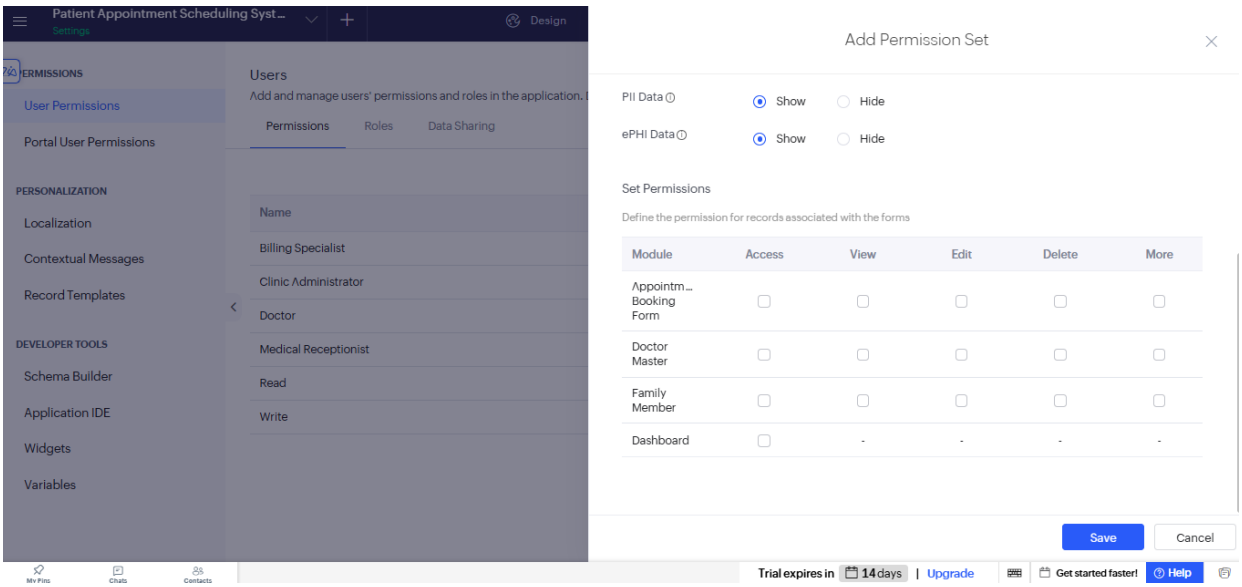


Figure 2: Add Permission Set dialog with module-level permission controls.

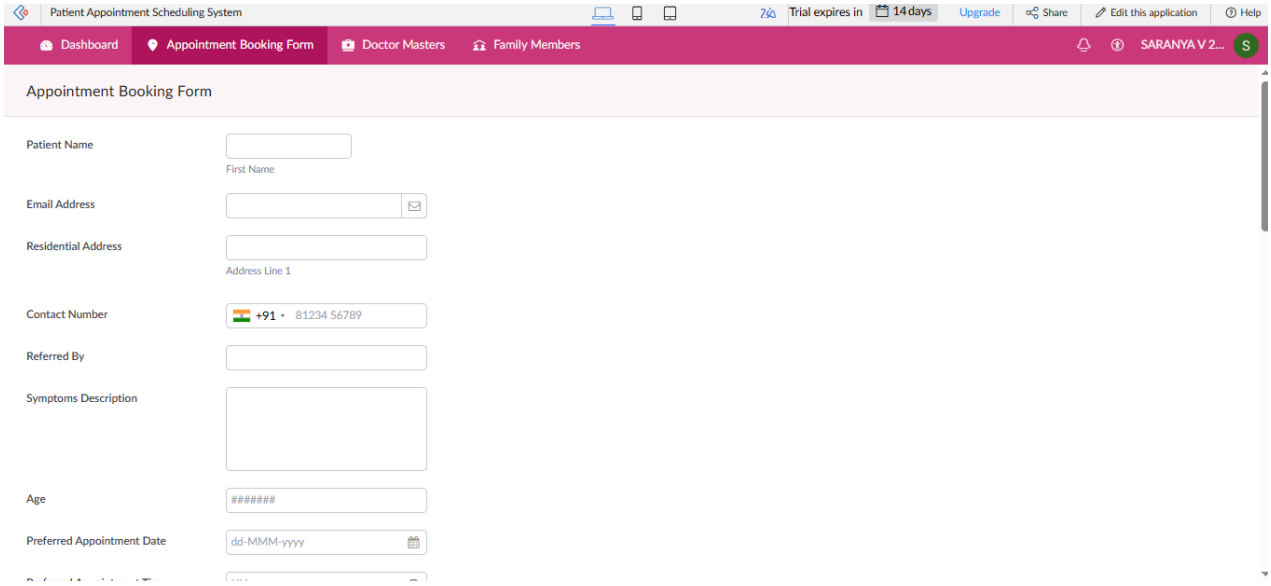


Figure 3: Appointment Booking Form used by the Patient Appointment Scheduling System.

7. Result

The User Permissions page and Add Permission Set controls were reviewed for the Patient Appointment Scheduling System. A role-based permission plan was prepared for clinic administrators, medical receptionists, doctors and billing specialists. The final access restrictions should be verified by assigning the sets to test users and checking the live application.

8. Learning Outcome

Learnt how permission sets, role hierarchy and data sharing can be used to implement RBAC in Zoho Creator, how module-level Access/View/Edit/Delete/More rights restrict functionality, and why the principle of least privilege is important for a patient appointment application.